Logarithmic convexity for a coupled system with two subdiffusion orders

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Abstract

We prove a logarithmic convexity estimate for a pair of coupled Caputo subdiffusion equations with different time orders. For every bounded smooth domain, every $0 < \alpha_1 < \alpha_2 < 1$, and every final time $T > 0$, the product L^2 norm at time t is bounded by a constant times the geometric interpolation of the initial and final norms, with exponent t/T. The estimate holds for arbitrary L^2 initial pairs and includes domains with growing low-frequency modes. This gives an affirmative answer to the Dirichlet norm-interpolation question for the system posed by Chorfi. The proof uses a positive matrix integral representation to establish finite-time invertibility and uniform bounds for all large spatial frequencies.

1 The problem and the main result

Subdiffusion equations describe evolution with a memory term in place of an ordinary time derivative. Their backward problem asks whether an earlier state can be recovered from a later observation. A logarithmic convexity estimate bounds an intermediate state by a geometric interpolation of the initial and final norms; in particular, it implies backward uniqueness. For a single Caputo order, Chorfi, Maniar and Yamamoto [2, Theorem 1.1] proved such an estimate for self-adjoint evolution equations. Coupled systems with a common time order admit a spectral treatment of their forward and backward problems [4]. Mixed-order coupled systems also have a well-developed mild-solution theory [3], but that existence theory does not itself give interpolation between the endpoint norms.

The distinction between a common order and two different orders is essential for the proof: a rotation that diagonalizes the reaction matrix mixes the two time derivatives. Chorfi [1, Section 5, problem 1] asks whether logarithmic convexity extends to the particular mixed-order system

$$\begin{aligned} D_t^{\alpha_1} u_1 &= \Delta u_1 + u_1 + u_2, \\ D_t^{\alpha_2} u_2 &= \Delta u_2 + u_1 + u_2. \end{aligned} \tag{1}$$

We consider homogeneous Dirichlet conditions on a bounded smooth domain $\Omega \subset \mathbb{R}^n$, with arbitrary initial pair $U_0 \in H := L^2(\Omega; \mathbb{C}^2)$ and $U = (u_1, u_2)$. The norm is $\|U\|^2 = \|u_1\|^2_{L^2(\Omega)} + \|u_2\|^2_{L^2(\Omega)}$. For an absolutely continuous scalar function, the Caputo convention is

$$D_t^{\alpha} h(t) = \frac{1}{\Gamma(1-\alpha)} \int_0^t (t-s)^{-\alpha} h'(s)\, ds.$$

For L^2 initial data, we use the continuous H representative of the componentwise mild solution, equivalently the fractional diffusion Duhamel formulation in [3, Definition 1]. The real case follows by restriction.

Theorem 1. *Let $\Omega \subset \mathbb{R}^n$ be any bounded smooth domain, $0 < \alpha_1 < \alpha_2 < 1$, and $T > 0$. There is a constant $C = C(\Omega, \alpha_1, \alpha_2, T) > 0$ such that every homogeneous Dirichlet mild solution of (1) with $U(0) = U_0 \in H$ satisfies*

$$\|U(t)\| \le C \|U_0\|^{1-t/T} \|U(T)\|^{t/T}, \qquad 0 < t < T. \tag{2}$$

The constant is independent of the initial pair and of the intermediate time. In particular, $U(T) = 0$ implies $U_0 = 0$.

This is the product-L^2 Dirichlet formulation of the question, including all initial pairs and the exact interpolation exponent t/T; it is also the formulation of AIM Problem 402.[1] No sign condition or additional initial regularity is required.

We keep the Dirichlet spatial eigenbasis and treat each pair of coefficients as a two-dimensional system. A positive matrix integral representation of its resolvent proves that the terminal map is invertible, including the finitely many modes that may grow. For large spatial eigenvalues, the same representation gives an upper smoothing bound and a matching lower bound for the smallest singular value at the final time. These bounds reduce (2) to scalar interpolation and Hölder's inequality. The constant is not asserted to be sharp or equal to one. The argument uses the common Laplacian and the symmetric constant coupling in (1); it does not treat general spatially varying coupling.

The modal representation and invertibility are proved in Section 2. Section 3 establishes the uniform estimates and proves the theorem. In the proof we write $a = \alpha_1$ and $b = \alpha_2$. Matrix norms and vector lengths are Euclidean; inequalities between real symmetric matrices refer to their quadratic forms.

2 The modal resolvent and finite-time invertibility

Let $0 < a < b < 1$ be fixed. A Dirichlet eigenvalue $\lambda > 0$ gives the two-dimensional system

$$D_t^a y_1 = -q y_1 + y_2, \qquad D_t^b y_2 = y_1 - q y_2, \qquad q = \lambda - 1 > -1. \tag{3}$$

Denote its fundamental matrix by $F_q(t)$, with $F_q(0) = I$, and put

$$A_q = \begin{pmatrix} q & -1 \\ -1 & q \end{pmatrix}, \qquad D(z) = \operatorname{diag}(z^a, z^b), \qquad R_q(z) = (D(z) + A_q)^{-1}.$$

All complex powers use the branch $\arg z \in (-\pi, \pi)$. The integral form of (3) is $y_i(t) = y_i(0) - I^{\alpha_i}(A_q y)_i(t)$, where $\alpha_1 = a$, $\alpha_2 = b$ and

$$I^\gamma f(t) = \frac{1}{\Gamma(\gamma)} \int_0^t (t-s)^{\gamma-1} f(s)\, ds.$$

It has a unique continuous solution of exponential order. Indeed, in the component maximum norm weighted by $e^{-\omega t}$, $\omega \ge 1$, its Volterra operator has norm at most $(|q| + 1)\omega^{-a}$. For large ω, the Neumann series converges. Local uniqueness also follows by iteration, since on $[0, T]$ the row kernels are bounded by $C_T t^{a-1}$ and their mth convolution integrals are bounded by $(C_T t^a)^m / \Gamma(1 + ma)$. Taking Laplace transforms gives

$$\widehat{F}_q(z) = \frac{1}{z} R_q(z) D(z) = \frac{1}{z}(I - R_q(z) A_q) \tag{4}$$

[1] The frozen problem statement is AIM Problem 402 at revision `a153f469ee0e`.

for positive real z sufficiently large. In particular, the different initial factors z^{a-1} and z^{b-1} have been retained.

The following positive matrix density will distinguish invertibility from mere positivity of the entries of a solution. For $r > 0$, define

$$\begin{aligned} M_q(r) &= A_q + \operatorname{diag}(r^a e^{i\pi a}, r^b e^{i\pi b}), \\ B(r) &= \operatorname{diag}(r^a \sin \pi a, r^b \sin \pi b), \\ W_q(r) &= -\frac{1}{\pi} \operatorname{Im} R_q(-r + i0) = \frac{1}{\pi} M_q(r)^{-*} B(r) M_q(r)^{-1}. \end{aligned}$$

Here X^* denotes conjugate transpose and $\operatorname{Im} X = (X - X^*)/(2i)$. The inverse identity $\operatorname{Im}(M^{-1}) = -M^{-*}(\operatorname{Im} M)M^{-1}$ proves the last equality. Since $B(r) > 0$, the density $W_q(r)$ is strictly positive definite. It is real symmetric: $R_q(z)$ is complex symmetric, so its Hermitian imaginary part is its entrywise imaginary part.

Lemma 2. *For every $q > -1$, the modal resolvent has the representation*

$$R_q(z) = \int_0^\infty \frac{W_q(r)}{z + r}\, dr + \begin{cases} K_q/(z - \rho_q), & -1 < q < 1, \\ 0, & q \geq 1. \end{cases} \tag{5}$$

In the first case, $\rho_q > 0$ is the unique positive pole and K_q is a positive semidefinite matrix of rank one. The integral is absolutely convergent away from the cut and the pole.

Proof. If $\Im z > 0$, then $\operatorname{Im} D(z) > 0$. Thus $D(z) + A_q$ has no kernel; the lower half-plane follows by conjugation. On the positive real axis $D(s) + A_q$ increases strictly in the positive definite ordering. Its eigenvalues start at $q-1, q+1$ and tend to positive infinity. It therefore has one zero, and only one, when $-1 < q < 1$, and none when $q \geq 1$. At the zero ρ_q, let v_q be a real unit kernel vector. The other eigenvalue is positive, and

$$K_q = \operatorname*{Res}_{z=\rho_q} R_q(z) = \frac{v_q v_q^T}{v_q^T D'(\rho_q) v_q}.$$

The denominator is positive, so this is a simple pole with the stated residue.

Apply the Cauchy formula entrywise to a keyhole contour about the negative real axis, retaining this pole when present. For fixed q, $R_q(z) = O(|z|^{-a})$ at infinity, uniformly in the argument, by factoring out $D(z)$. It is bounded at zero if $q \neq 1$, since $q = -1$ is excluded. At $q = 1$ its explicit inverse is

$$R_1(z) = \frac{1}{z^{a+b} + z^a + z^b} \begin{pmatrix} z^b + 1 & 1 \\ 1 & z^a + 1 \end{pmatrix} = O(|z|^{-a}) \quad (z \to 0). \tag{6}$$

The small and large contour integrals consequently vanish. The jump between the two banks of the cut is $-2\pi i W_q(r)$, giving (5) with the displayed residue sign. At infinity $W_q(r) = O(r^{-a})$; at zero it is $O(r^a)$ if $q \neq 1$ and $O(r^{-a})$ if $q = 1$. These bounds also prove the asserted absolute convergence. No bound uniform in q is needed in this lemma. □

Define the time kernel and its integral by

$$Q_q(t) = \int_0^\infty e^{-rt} W_q(r)\, dr + \begin{cases} e^{\rho_q t} K_q, & q < 1, \\ 0, & q \geq 1, \end{cases} \qquad G_q(t) = \int_0^t Q_q(s)\, ds. \tag{7}$$

The kernel is strictly positive definite. Its singularity at zero is at most $C_q t^{a-1}$, so it is locally integrable. Absolute Fubini and (5) identify its Laplace transform with R_q to the right of the pole. Equation (4) and uniqueness of the Laplace transform therefore yield

$$F_q(t) = I - G_q(t) A_q. \tag{8}$$

Proposition 3. *The matrix $F_q(t)$ is invertible for every $q > -1$ and every finite $t \geq 0$.*

Proof. Suppose first that $q \neq 1$. The endpoint bounds in Lemma 2 ensure convergence of

$$P_q(t) = \int_0^\infty e^{-rt} \frac{W_q(r)}{r}\, dr > 0 \qquad (t \geq 0).$$

For $q < 1$, evaluating (5) at zero and integrating (7) give, respectively,

$$A_q^{-1} = P_q(0) - K_q/\rho_q,$$
$$G_q(t) = P_q(0) - P_q(t) + (e^{\rho_q t} - 1)K_q/\rho_q.$$

For $q > 1$, the same identities hold with the pole terms omitted. Their constant terms cancel in (8), giving the real symmetric matrix

$$H_q(t) := F_q(t)A_q^{-1} = P_q(t) - \begin{cases} e^{\rho_q t}K_q/\rho_q, & q < 1, \\ 0, & q > 1. \end{cases} \tag{9}$$

For $q > 1$, both $H_q(t)$ and A_q are positive definite, and the result follows. For $-1 < q < 1$, the matrix A_q^{-1} has one eigenvalue of each sign. Since $H_q(t) = A_q^{-1} - G_q(t)$, its negative direction remains strictly negative. A nonzero vector in $\ker K_q$ has strictly positive quadratic form under $H_q(t)$, by (9). Hence $H_q(t)$ has one eigenvalue of each sign and is nonsingular.

For $q = 1$, put $w = (1, -1)^T$, so $A_1 = ww^T$. From (6),

$$w^T R_1(s) w = \frac{s^a + s^b}{s^{a+b} + s^a + s^b} \longrightarrow 1 \qquad (s \downarrow 0).$$

Writing $k(r) = w^T W_1(r) w > 0$, the representation (5) and monotone convergence imply $\int_0^\infty k(r)/r\, dr = 1$. The rank-one determinant formula in (8) now gives

$$\det F_1(t) = 1 - w^T G_1(t) w = \int_0^\infty e^{-rt} \frac{k(r)}{r}\, dr > 0.$$

This also covers a domain having the critical Dirichlet eigenvalue $\lambda = 2$. At $t = 0$, invertibility is immediate from $F_q(0) = I$. □

3 Uniform estimates and the interpolation inequality

Invertibility of each fixed mode alone is insufficient for the desired estimate. We next control all large eigenvalues with the same constants. Throughout this section, $T > 0$ remains fixed.

Lemma 4. *There are $M = M(a, b, T) > 0$ and $d = d(a, b, T) > 0$ such that, for every $q \geq 2$,*

$$\|F_q(t)\| \leq \frac{M}{1 + qt^b} \quad (0 < t \leq T), \qquad |F_q(T)v| \geq \frac{d}{\sqrt{3q}}\, |v| \quad (v \in \mathbb{C}^2). \tag{10}$$

Proof. The eigenvalues of A_q are $q - 1, q + 1$, so its condition number is at most three. The positive integral in (9) gives

$$S_q(t) := A_q^{1/2} H_q(t) A_q^{1/2} = A_q^{1/2} F_q(t) A_q^{-1/2}.$$

This is positive definite, decreases in the ordering of symmetric matrices, and satisfies $S_q(0) = I$. For $s > 0$, set $E = A_q^{-1/2} D(s) A_q^{-1/2} > 0$. Equation (4) yields

$$\widehat{S}_q(s) = \frac{1}{s}(I - (I + E)^{-1}) = \frac{1}{s} E(I + E)^{-1} \leq \frac{\max(s^a, s^b)}{(q - 1)s} I.$$

On the other hand, monotonicity gives

$$\widehat{S}_q(1/t) \geq \int_0^t e^{-s/t} S_q(t)\, ds = t(1-e^{-1})S_q(t).$$

Since also $S_q(t) \leq I$, returning through the similarity gives

$$\|F_q(t)\| \leq \sqrt{3}\min\left(1, \frac{\max(t^{-a}, t^{-b})}{(1-e^{-1})(q-1)}\right). \tag{11}$$

Let $L_T = \max(1, T^{b-a})$. For $0 < t \leq T$ we have $\max(t^{-a}, t^{-b}) \leq L_T t^{-b}$, and $q - 1 \geq q/2$. Using $\min(1, c/x) \leq (1+c)/(1+x)$ for $c, x > 0$ in (11) proves the first estimate, for example with $M = \sqrt{3}(1 + 2L_T/(1-e^{-1}))$.

For the lower estimate, recall the positive density

$$W_q(r) = \frac{1}{\pi} M_q(r)^{-*} B(r) M_q(r)^{-1}.$$

Here the factors are

$$M_q(r) = A_q + \operatorname{diag}(r^a e^{i\pi a}, r^b e^{i\pi b}),$$
$$B(r) = \operatorname{diag}(r^a \sin \pi a, r^b \sin \pi b).$$

Set $\eta = \min(\sin \pi a, \sin \pi b) > 0$ and $L = 1 + 2^b$. For $1 \leq r \leq 2$, $B(r) \geq \eta I$ and $\|M_q(r)\| \leq q + L$. Therefore

$$W_q(r) \geq \frac{\eta}{\pi(q+L)^2} I.$$

Restricting the positive integral for $H_q(T)$ to this interval gives

$$S_q(T) \geq \frac{\eta e^{-2T}(q-1)}{2\pi(q+L)^2} I \geq \frac{d}{q} I, \qquad d = \frac{\eta e^{-2T}}{4\pi(1+L/2)^2}.$$

Here $q - 1 \geq q/2$ and $q + L \leq q(1 + L/2)$ were used. Since $F_q(T) = A_q^{-1/2} S_q(T) A_q^{1/2}$, its smallest singular value is at least $(d/q)\sqrt{(q-1)/(q+1)} \geq d/(\sqrt{3}q)$. □

We now deduce the modal interpolation estimate. Put $\theta = t/T$ and

$$h_q = (1 + qT^b)^{-1}, \qquad m = \min(1, dT^b/\sqrt{3}) > 0.$$

Lemma 4 implies $|F_q(T)v| \geq m h_q |v|$ for $q \geq 2$. The function $-\log(1 + qt^b)$ is convex on $[0, T]$ and vanishes at zero. Consequently,

$$(1 + qt^b)^{-1} \leq h_q^\theta,$$

and the same lemma gives

$$|F_q(t)v| \leq \frac{M}{m} |v|^{1-\theta} |F_q(T)v|^\theta \qquad (q \geq 2,\ 0 < t < T). \tag{12}$$

The zero vector is included directly. No simultaneous diagonalization of the two time derivatives is used.

Proof of Theorem 1. Let (λ_j, ϕ_j) be an orthonormal Dirichlet eigensystem of $-\Delta$, and set $q_j = \lambda_j - 1$ and $v_j = ((u_1(0), \phi_j), (u_2(0), \phi_j))^T$. Projection of the componentwise mild equations gives (3) in its fractional integral form. Thus its uniqueness gives

$$U(t) = \sum_{j=1}^{\infty} \phi_j F_{q_j}(t) v_j. \tag{13}$$

This uses only a bounded spectral coefficient functional on the mild integrals. It does not apply Δ to U_0. Equivalently one can start with the mild formulation in [3, Definition 1 and Theorem 1] and project it. Lemma 4 and the finitely many lower modes give a uniform operator bound on $[0,T]$; hence the series converges in product L^2 and is continuous there, including at $t=0$, by dominated convergence.

There are only finitely many indices with $q_j < 2$. If there are any, define

$$M_0 = \max_{q_j<2} \max_{0\le s\le T} \left\| F_{q_j}(s) \right\|, \qquad m_0 = \min_{q_j<2} \sigma_{\min}(F_{q_j}(T)).$$

Continuity and Proposition 3 give $M_0 < \infty$ and $m_0 > 0$. These modes satisfy (12) with constant $M_0/\min(1,m_0)$, since $m_0^\theta \ge \min(1,m_0)$. Let C be the larger of this constant and M/m, omitting the first constant if the low-frequency set is empty. Then C depends only on Ω, a, b, T, and Parseval and Hölder give

$$\begin{aligned} \|U(t)\|^2 &\le C^2 \sum_j |v_j|^{2(1-\theta)} |F_{q_j}(T)v_j|^{2\theta} \\ &\le C^2 \left(\sum_j |v_j|^2 \right)^{1-\theta} \left(\sum_j |F_{q_j}(T)v_j|^2 \right)^{\theta} \\ &= C^2 \|U_0\|^{2(1-\theta)} \|U(T)\|^{2\theta}. \end{aligned}$$

Taking square roots proves the result for every U_0 and $0 < t < T$. If $U(T) = 0$, Proposition 3 shows that every $v_j = 0$, so the same conclusion holds without a division by zero. □

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References

[1] S.-E. Chorfi, *Logarithmic convexity of evolution equations and application to inverse problems*, arXiv:2506.19954v1 (2025), arXiv:2506.19954v1.

[2] S. E. Chorfi, L. Maniar and M. Yamamoto, *The backward problem for time fractional evolution equations*, arXiv:2211.16493v1 (2022), arXiv:2211.16493v1.

[3] Z. Li, X. Huang and Y. Liu, *Initial-boundary value problems for coupled systems of time-fractional diffusion equations*, Fractional Calculus and Applied Analysis **26** (2023), 533–566, doi:10.1007/s13540-023-00149-0.

[4] D. Feng, Y. Liu and S. Lu, *Forward and backward problems for coupled subdiffusion systems*, arXiv:2407.00588v2 (2025), arXiv:2407.00588v2.